

Project Proposal (Week 2), Energy-Aware Ventilation & Climate Control

D7065E, worked example

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About this document. A short, worked example of the **Week 2 proposal**. It is deliberately provisional, its job is to get a *direction* approved, not to lock down the design. Some early choices here (REST transport, an ML occupancy model) are revised later in the final report; that is expected and good.

Summary

Team	A. Andersson, B. Berg
Use case	Energy-aware ventilation & climate control
Pitch	Ventilate and heat rooms based on who is actually present, to keep air quality and comfort high while cutting energy.
Repository	github.com/example-team/d7065e-ventilation

1. Use case, and why

This project controls ventilation and heating in a BuildSim office floor. It was chosen because it has a genuine tension, air quality and comfort pull one way, energy the other, which makes the autonomous decision interesting, and because BuildSim provides rooms and occupancy to work with.

2. Scenario

A meeting fills a room; CO₂ and temperature rise; the system detects it and increases ventilation, holding CO₂ below 1000 ppm; when the room empties it falls back to a low-energy setback. The same loop runs in every room.

3. What will be built (*rough sketch*)

- **Sensors:** temperature, CO₂, and occupancy per room, registered with BuildSim.
- **Autonomous decision-maker:** sets ventilation level and heating setpoint per room.
- **Actuators:** HVAC setpoint and ventilation damper, writing state back to BuildSim.
- **Data pipeline / storage:** collect sensor readings, store recent history, and provide data to the autonomous service and the dashboard.
- **Dashboard:** overlay state and decisions on the BuildSim 3D view.

Each major component (sensors, autonomous service, actuators, storage, and dashboard) runs as a **separate process** communicating through **defined interfaces**, and talks to BuildSim through its API.

4. Simulating the building

BuildSim provides the rooms; the physics and the people are simulated as follows.

- **Physical processes:** a simple per-room model, temperature from a heat balance (envelope loss, body heat, HVAC), and CO₂ that rises with the number of people and falls with ventilation. Actuator commands feed back into the model so the loop closes.
- **People movement (occupancy):** sample a head-count per room conditioned on **room type and time of day**, offices steady through the day, the fika room busy around 11:30, conference rooms full during meetings, empty at night. (A richer option that could be added later: move individual people between rooms along BuildSim's **navigation graph**.)

The models are intentionally simple, but **actuator actions must influence future sensor readings** so that the control loop closes.

5. The autonomous part

It decides each room's fan level and setpoint from CO₂, temperature, and occupancy. **Starting approach:** a small ML model to predict occupancy, feeding a simple controller. Plain rules may be used instead if the data turns out too thin to train on.

6. Biggest unknowns & risks

- Whether two weeks of simulated data is enough to train the occupancy model.
- Whether sending readings to the controller over REST keeps up as rooms are added.
- Whether the room physics is realistic enough for the decisions to mean anything.

7. First step & rough plan

The first step is **one room working end-to-end**, one set of sensors → controller → actuator → BuildSim, to prove the loop closes. After that, more rooms, storage, and the dashboard are added, and failure handling and evaluation follow toward the Week 8 demo.